COM4302 Computer Science Fundamentals

Assessment 001 brief

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Assessment: 001 Portfolio: A portfolio of different academic elements. This may include weekly class activities, in class tests, presentations, assignments and self-directed learning activities.

Due Date: 16/01/2026

Weighting: 100%

Assessment brief

Several activities will be assigned to the students, including homework programming exercises, keeping a logbook, in-class quizzes, and 2 programming assignments, aiming to understand the fundamental Computer Science concepts and principles, a short presentation of their 2nd assignment, and a short essay on professional issues and practice.

A. Homework exercises logbook

You will be assigned weekly programming exercises. One of them will be specified as "homework"
and your solution, along with any comments on design of the solution, error handling, etc., should be kept in a logbook.
A template for the logbook will be provided.

B. Programming assignment 1

Implement a simple game
Scenario: Implement the Rock-Paper-Scissors game. The user of the application will play against the computer.
Initially, the user types a username and the game starts.
The user selects one of the three options, and the computer selects randomly one, too.
Then, the program examines both options and announces the winner, or indicates tie in case both, the user and the computer, selected the same option. Also, the program updates the score.
The game continues until one of the players, user or the computer, gets a score of 10.
Finally, the score is recorded in a text file as:
username,datetime,user_score,computer_score.
Write another program to display the top 5 players.

C. Programming assignment 2

Implement a Quiz application

Scenario: The application will use a file with the users, containing username and password. On starting up the user will have to login. If the user has not an account an option to register will be provided.

After login, the program will use a file with multiple choice questions(*) to select 5 questions randomly. Alternatively, the questions may be retrieved from the web (API URL will be provided).

The program will display each question to the user, along with the 4 answers and prompt the user to respond with the correct answer.

Then, the program will check if the user responded correctly and update the score.

In the end, a score file will be updated and the user will be asked if they want to play again.

(*) file format: question,correct_answer,wrong_ans1,wrong_ans2,wrong_ans3

D. In class tests

There will be 2 class tests with a number of programming questions.

E. Report on professional issues

Write a short report of 800-1000 words. In this report investigate the opportunities computing studies give, and reflect on your personal plans and ambitions, based on your findings.

Deliverables and successful submission

1: A .zip file with the logbook, and solutions to the programming assignments (upload on the e-learning platform OR provide a link to your GitHub project(s))

2: Report (upload on Turnitin)

Grading scheme

Topic	Marks
A. Homework exercises logbook	20%
B. Programming assignment 1	20%
C. Programming assignment 2	30%
D. In class tests	15%
E. Report on professional issues	15%

Use of Generative Artificial Intelligence (GAI) Applications in this Assessment

AI Status	Application	Notes
Category C	GAI is an integral part of the assessment and should be used.	Grammar and/or spell checkers may be used to correct individual words and sentences. Your skills in using and evaluating GAI generated content is part of the assessment task in question. GAI generated content should be clearly identified and acknowledged. Any GAI generated content which is presented as your own original work and is not acknowledged will be assessed for academic misconduct.

References and citation

Written work should be referenced using the standard University of Bolton referencing style (Harvard Referencing Style)– see:
https://www.bolton.ac.uk/library/Study-Skills/Referencing/Home.aspx

Minimum Secondary Research Source Requirements:

Level HE4 - It is expected that the Reference List will contain between **five and ten sources**. As a MINIMUM the Reference List should include **one refereed academic journal** and **three academic books**

Rationale Report Criteria

Relevance 15%
Knowledge 20%
Argument/Analysis 20%
Structure 10 %
Presentation 15%
Written English 10%
Research/Referencing 10%

Declaration: At the end of the assessment, you should also include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment.

Late Submissions and Plagiarism

Late submissions will be handled according to the University's Assessment Regulations for Undergraduate Programmes (see https://www.bolton.ac.uk/wp-content/uploads/2018/11/Assessment-Regulations-for-Undergraduate-Programmes-2018-2019.pdf).

Plagiarism (cheating) is a serious academic offence.

Guidance about and to avoid plagiarism can be found on the University's LEAP Online website (see https://www.bolton.ac.uk/leaponline/My-Academic-Development/My-Writing-Techniques/Plagiarism.aspx).

Module Learning Outcomes

Learning Outcomes	**Assessment**
LO1: Evaluate career paths in Computing and consider issues relating to professional practice	YES
LO2: Design and construct solutions to a derived specification using fundamental Computer Science concepts and principles	YES